

Testing and Validation Summary (Milestone 3)

The MVP DAO platform has undergone structured functional, security, and scalability testing to validate core system behaviour under real-world conditions.

Functional validation was conducted using an end-to-end automated testing suite, simulating key user flows including registration, contribution completion, campaign interaction, and on-chain reward distribution. All core flows executed successfully on Cardano pre-production infrastructure, confirming that the system supports intended user actions and token mechanics.

Scalability testing was performed at the application and blockchain interaction layers. Load testing of the API layer demonstrated stable performance under concurrent usage, with consistent response times and no request failures. Transaction throughput testing confirmed successful on-chain execution of reward distribution flows, including improvements following optimisation of UTxO handling.

Security testing included dependency auditing, access control review, and identification of potential vulnerabilities across authentication and data layers. Known issues have been documented and prioritised, with critical and high-severity items clearly identified for resolution in subsequent development phases.

These results demonstrate that the MVP platform is functionally operational, capable of supporting core governance and participation flows, and has been tested against realistic usage scenarios. The current testing phase establishes a validated baseline for further security hardening, scalability optimisation, and feature refinement.

A roadmap for issue resolution and re-testing has been defined, with subsequent phases focused on addressing identified vulnerabilities, strengthening access controls, and preparing the platform for broader deployment. The current testing phase establishes a validated MVP baseline, with pathways toward production-grade security and scalability in the next phases.